SUPPLEMENT I—Joint World Bank/IMF Debt Sustainability Analysis

This debt sustainability analysis (DSA), prepared jointly by the staffs of the IMF and the World Bank, applies the low-income country template to the staff's projections under a no-reform scenario and a reform scenario, the latter also factoring in potential debt relief under the Enhanced HIPC Initiative and MDRI. The analysis shows that Togo is in debt distress, with large arrears to the World Bank and other official creditors and NPV-of-debt ratios projected to remain well above the country-specific thresholds over the long term. The reform scenario illustrates the potential gains associated with possible debt relief, fiscal adjustment, and economic reforms. Different stress tests indicate that public and external debt would remain sustainable under the reform scenario; in the absence of fiscal and structural reforms, however, key debt indicators would revert to an unsustainable path.

Introduction

1. Togo is currently participating in a staff-monitored program (SMP) with the IMF, which could lead to a Fund-supported program under the Poverty-Reduction and Growth Facility (PRGF) and, eventually, debt relief under the Enhanced HIPC Initiative and MDRI.[1] The staffs have therefore prepared two scenarios, a no-reform scenario characterized by low growth and high fiscal deficits, and a reform scenario describing an economic recovery supported by structural reforms, higher external assistance, and potential debt relief. The former scenario is used to determine the extent of Togo's debt burden. The latter scenario illustrates the potential impact of reforms and debt relief and is used to discuss the robustness of the projected debt indicators (post-debt relief) to various shocks.

Current debt situation

2. **Togo's public sector debt, including arrears, was estimated at US$2.3 billion as of end-2006, corresponding to 103 percent of GDP.** External debt was estimated at US$ 1.8 billion (81 percent of GDP), of which the World Bank and Paris Club creditors account for the lion's share, with about US$0.7 billion each. Arrears now account for almost a third of total external debt. Togo's debt to the IMF is relatively modest (US$8.2 million as of end-2006), and it is current on this debt. While public debt has been declining since 2000, this occurred mainly because the CFA franc (pegged to the euro) has appreciated against the

[1] Togo belongs to the "ring-fenced" countries that met the end-2004 income and indebtedness criteria under the Enhanced HIPC Initiative and would potentially be eligible for debt relief under the Enhanced HIPC Initiative and MDRI. With domestic revenues exceeding 15 percent of GDP, and an export-GDP ratio exceeding 30 percent, Togo would qualify for debt relief under the Enhanced HIPC Initiative under the "fiscal window;" and a NPV of debt-to-revenue threshold of 250 percent would apply.

U.S. dollar and other currencies.[2] Without this appreciation, debt-to-GDP ratio would have been about 20 percent above its current level.

Figure 1. The most important creditors are the World Bank and the Paris Club.

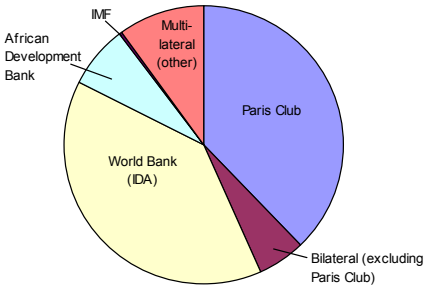
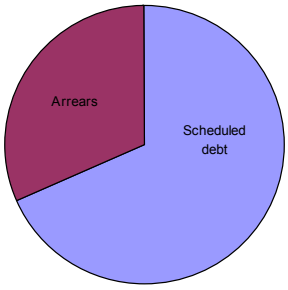


Figure 2. About one-third of Togo's external debt is in arrears.



3. **Togo is in debt distress**. Togo is in arrears to the World Bank, the African Development Bank, and other multilateral and bilateral creditors (but not the IMF). At end-2006, the net present value (NPV) of external debt to exports ratio was estimated at 162 percent, and the ratio of the NPV of debt to revenue at 388 percent, in both cases well above the applicable indicative thresholds (Table 1). Moreover, because Togo has not serviced most of its external debt in recent years, arrears on principal and interest have accumulated rapidly and now account for a substantial proportion of overall debt (Figure 2). The indicators of debt service relative to exports or domestic revenues are below the applicable thresholds, partly because a substantial proportion of debt is in arrears.[3]

[2] Between end-2000 and end-2006, the CFA franc appreciated against the U.S. dollar by 42 percent, from 729 CFA francs per U.S. dollar to 513 CFA francs per U.S. dollar.

[3] If an imputation for repayment of arrears is included, assuming a 20-year period repayment period and an interest rate of 4 percent, debt service would amount to 11 percent of exports and 26 percent of government revenues (excluding grants) in 2006, the latter exceeding the applicable indicative threshold.

Table 1. Togo: Key External Debt Sustainability Indicators (Percent)[1]
(No-reform scenario)

	Indicative Threshold[2]	2006 (Est.)	2027 (Proj.)
NPV of debt to GDP	30	65.6	52.6
NPV of debt to exports	100	162.1	133.3
NPV of debt to revenue (excl. grants)	200	387.9	329.5
Debt service to exports	15	7.1	8.3
Debt service to revenue (excl. grants)	25	16.9	20.4

[1] All debt indicators refer to the NPV of public and publicly guaranteed external debt, and relate to the"no-reform scenario" described in Table 2.
[2] Policy-dependent debt-sustainability threshold over which countries with similar evaluations of policies and institutions would face a probability of about 20 percent of experiencing a prolonged incident of debt distress in the coming year. The thresholds shown relate to low-income countries with poor institutional quality, reflecting Togo's score in the World Bank's Country Policy and Institutional Assessment (CPIA, average 2003–05), which lies within the lowest quintile among the countries covered.

Long-term economic macroeconomic outlook

4. **Togo's growth performance has been very weak, with per capita income declining since the early 1980s.** Political turmoil and weak governance have undermined the investment climate, while access to credit has been hampered by the poor performance of state-owned banks. In the absence of comprehensive economic reforms and increased external assistance, this is unlikely to change. The no-reform scenario (illustrated in Table 2) therefore projects GDP growth to hover around its long-term average of 1¾ percent (below population growth), an overall fiscal deficit exceeding 5 percent of GDP, continued accumulation of arrears, and large external imbalances. In this scenario, debt indicators related to the NPV of debt remain well above the indicative thresholds shown in Table 1 and Figures 6–7. Debt sustainability indicators related to the debt service due remain below the applicable thresholds (see, for example, Fig. 8), largely because a substantial proportion of debt is in arrears, with no debt service scheduled.

5. **The more favorable outlook in the reform scenario (Table 2) is based on the implementation of a comprehensive reform program, coupled with increased external assistance and the estimated impact of potential debt relief through the Enhanced HIPC Initiative and MDRI.** Real GDP growth rises to around 3½-4 percent per year, sustained by an increase in foreign direct investment, greater availability of credit to the private sector, and upgrades to critical infrastructure. To achieve this, the scenario envisages an increase in external assistance to 6 percent of GDP (with two-thirds in grants), steps to improve efficiency through improved governance, reallocation of fiscal spending toward public investment; steps to improve the business environment; reform of state-owned enterprises; and banking sector reform.

Table 2. Togo: Macroeconomic Outlook Under the No-Reform and Reform Scenarios
(Percent of GDP, unless otherwise indicated)

	No-Reform Scenario			Reform Scenario		
	2007	2017	2027	2007	2017	2027
Investment and savings						
Real GDP (percent change)	2.1	1.7	1.7	2.9	3.5	3.5
Real GDP per capita (percent change)	0.4	0.9	0.9	1.2	2.6	2.6
GDP (CFAF billions)	1,206.8	1,707.4	2,415.6	1,216.3	2,163.2	3,719.6
Investment and savings						
Gross domestic investment	12.5	12.3	12.3	13.7	22.6	22.6
Government	3.4	3.3	3.3	3.6	7.0	7.0
Nongovernment	9.1	9.0	9.0	10.1	15.6	15.6
Gross national savings	6.1	4.7	4.6	7.3	18.1	18.2
Government	1.1	-1.9	-2.1	1.1	5.5	5.6
Nongovernment	5.0	6.5	6.7	6.3	12.6	12.6
Government budget						
Total revenue and grants	17.5	17.0	17.0	19.0	22.4	22.4
of which: grants	1.6	1.0	1.0	2.6	4.2	4.2
Total expenditure and net lending	21.1	22.1	22.4	21.6	23.9	23.8
Overall balance (payment order basis)	-3.6	-5.2	-5.4	-2.5	-1.5	-1.4
Domestic primary balance	-1.4	-2.2	-2.2	-0.4	1.0	1.0
External sector						
Current account balance	-6.4	-7.6	-7.7	-6.4	-4.6	-4.4
Foreign direct investment	2.5	3.0	3.0	2.5	4.5	4.5

Sources: Togolese authorities; and Fund staff estimates and projections.

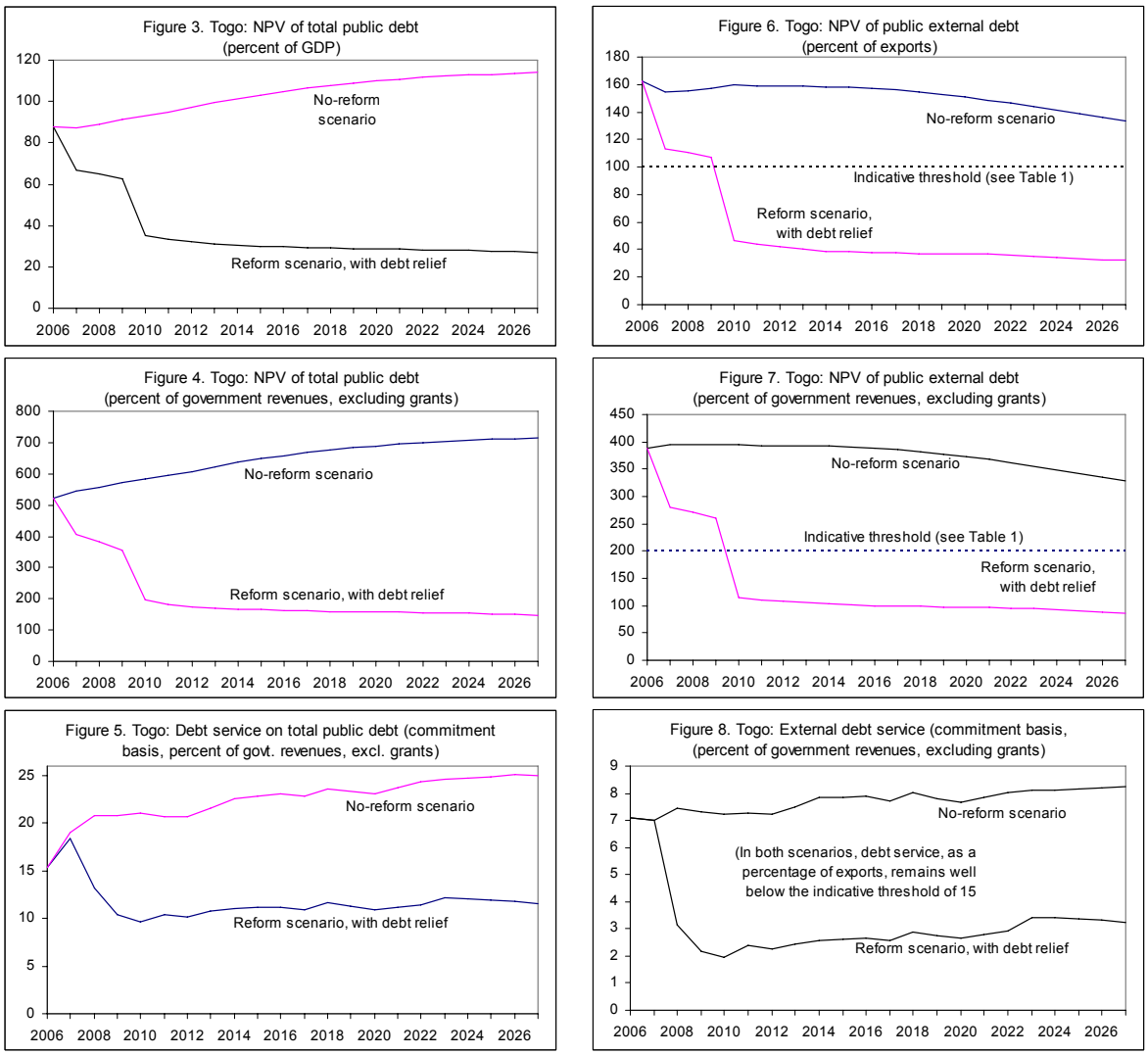
6. **As a prerequisite for a PRGF arrangement, Togo would have to reach understandings on the clearance of outstanding arrears on official external debt.** In this regard, the EU is preparing several large grants that could be used in part to clear Togo's arrears to the European Investment Bank (about $20 million). The World Bank staff is preparing a development policy needs review and is examining options for clearing arrears, which stood at about $105 million at end-2006. The authorities intend to intensify their dialogue with multilateral creditors and approach the Paris Club to request a rescheduling of bilateral debt and arrears.

Debt Sustainability Analysis

7. **Projections of public debt and debt service under the no-reform scenario show that Togo's public debt is unsustainable in the absence of policy adjustments and debt relief** (Figures 3-8, and Tables 3-4). In this scenario, the NPV of total public debt (external and domestic) is projected to rise from 88 percent of GDP in 2006 to 114 percent of GDP in 2027, and from 520 percent of government revenues (excluding grants) to 714 percent (Figures 3–4). Debt service would rise from 15 percent of government revenues (excluding grants) to 25 percent over the same period (Figure 5). A similar picture emerges for external debt. While the NPV of external debt would decline slowly from 388 percent of government

revenues in 2006 to 329 percent in 2027, it would remain well above the indicative threshold of 200 percent, as illustrated in Figures 6–7. The ratio of the NPV of external debt to exports would also remain above the indicative threshold of 100 percent (falling from 162 percent in 2006 to 133 percent in 2027), while external debt service would increase somewhat (to 8 percent of exports, by 2027).

Figures 3–8. In the no-reform scenario, debt indicators remain well above the applicable thresholds, while the reform scenario with debt relief envisages a very substantial decline.



8. **The reform scenario projects a steep decline in external and total public debt,** owing to debt relief in the context of the Enhanced HIPC Initiative and MDRI, but also an improved macroeconomic and fiscal outlook in the longer run, owing to the implementation of a comprehensive reform program. In this scenario, the NPV of total public debt is projected to decline to 27 percent of GDP (148 percent of government revenues) by 2027, while the NPV of external debt falls to 16 percent of GDP, or 32 percent of exports and 85 percent of government revenues (Table 5). External debt service declines steeply owing to

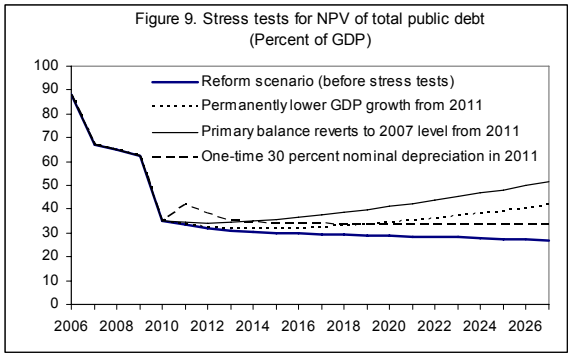
debt and arrears rescheduling and cancellation, including under HIPC/MDRI, between 2007 and 2010, and then increases modestly to around 3 percent of exports (Table 7). All debt indicators remain substantially below the applicable debt thresholds shown in Table 1.

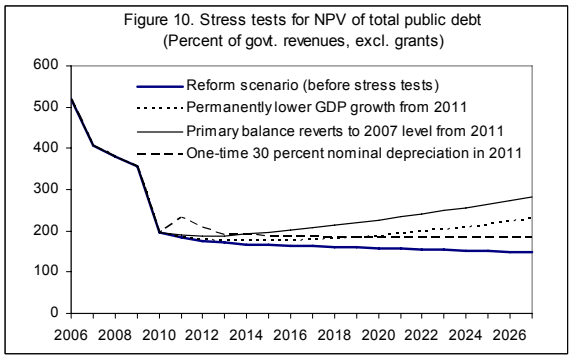
9. **Stress tests focus on debt sustainability and the robustness of the key debt indicators under the reform scenario after debt relief.** All shocks are thus assumed to occur from 2011 on. Stress tests are summarized in Figures 9–11 and Table 6 (for total public debt), and in Figures 12–14 and Table 8 (for external debt).

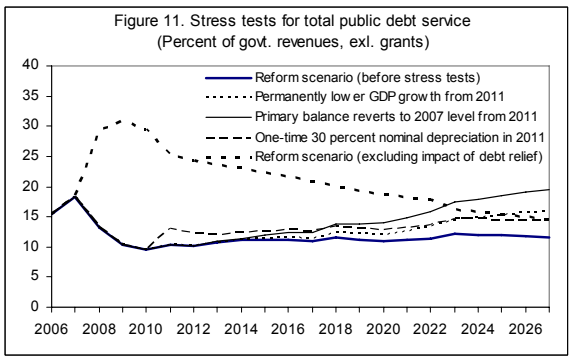
10. **Debt indicators remain below the applicable thresholds under most stress tests that simulate temporary shocks or policy reversals.** Under most temporary shocks and reversals, debt increases in 2011–12 and gradually declines thereafter, reaching a sustainable level in the long term (see bound tests in Tables 6 and 8). External debt is particularly sensitive to stress test B4 (Table 8), which envisages a debt-financed shortfall in private transfers from abroad and FDI totaling about 20 percent of GDP spread over two years. In this case, the NPV of external debt rises above the relevant indicative threshold (Figure 12), but eventually reverts to a sustainable level. Under a severe output shock (bound test B1 of Table 6), with two successive years of real GDP growth set at 5½ percentage points below trend, public debt as a share of GDP would increase and would not revert to a lower level.

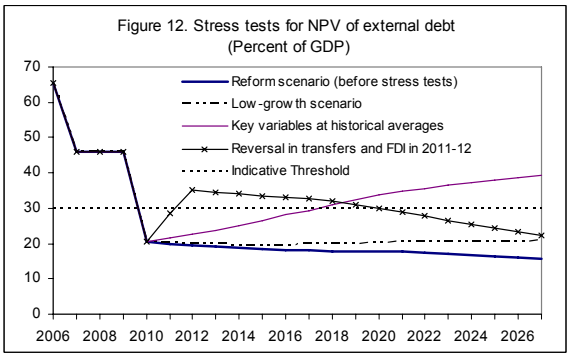
11. **Debt may revert to an unsustainable path (even after debt relief) in case of a permanent reversal in fiscal/economic reforms or a permanent economic shock without an adequate policy response.** As shown in the alternative scenarios in Tables 6 and 8 (Figures 9-14), if key variables (such as growth, fiscal balances, external flows) revert to their historical averages, debt will generally increase over time and, in some cases, exceed indicative thresholds by 2027. Similarly, bound test B5 in Table 6 shows that a one-off increase in public debt by 30 percent would result in a gradually rising level of debt over the long term. This underlines that debt relief alone is insufficient to ensure debt sustainability if economic and fiscal reforms are not sustained. It also shows that permanent economic shocks will require fiscal adjustment beyond the policies assumed in the reform scenario to ensure debt sustainability.

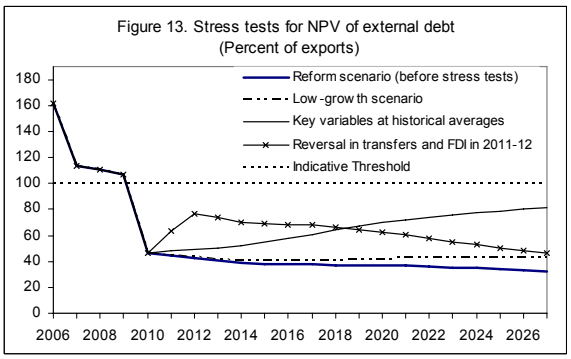
Figures 9–14. Stress Tests Under the Reform Scenario with Debt Relief.











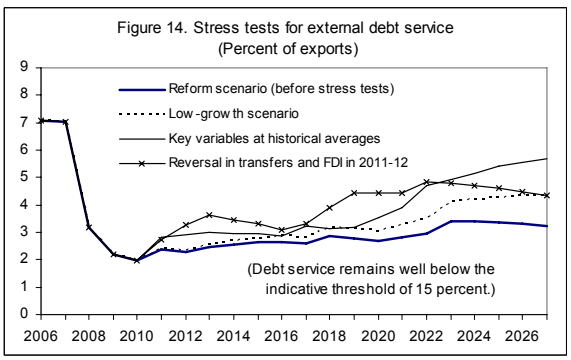


Table 3. Togo: Public Sector Debt Sustainability Framework, No-Reform Scenario, 2004-2027
(In percent of GDP, unless otherwise indicated)

	Actual			Projections									
	2006	Historical Average[5]	Standard Deviation[5]	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt[1]	102.9			103.3	104.9	107.0	109.0	110.8	113.1		123.6	141.8	
o/w foreign-currency denominated	80.5			79.3	78.9	78.8	78.7	78.6	78.6		78.6	80.5	
Change in public sector debt	1.3			0.4	1.7	2.1	2.0	1.8	2.3		2.0	1.7	
Identified debt-creating flows	-7.1			-1.9	0.0	0.6	0.6	0.3	1.3		1.0	0.7	
Primary deficit	1.9	-0.1	2.4	2.6	3.5	3.7	3.7	3.7	3.7	3.5	3.7	3.7	3.7
Revenue and grants	18.3			17.5	17.0	17.0	17.0	17.0	17.0		17.0	17.0	
of which: grants	1.4			1.6	1.0	1.0	1.0	1.0	1.0		1.0	1.0	
Primary (noninterest) expenditure	20.2			20.2	20.5	20.7	20.7	20.7	20.7		20.7	20.7	
Automatic debt dynamics	-8.9			-4.5	-3.5	-3.1	-3.2	-3.4	-2.5		-2.7	-3.1	
Contribution from interest rate/growth differential	-3.1			-3.2	-2.4	-2.4	-2.4	-2.6	-2.6		-2.9	-3.2	
of which: contribution from average real interest rate	-1.1			-1.1	-0.7	-0.6	-0.6	-0.7	-0.8		-0.8	-0.9	
of which: contribution from real GDP growth	-2.0			-2.1	-1.7	-1.8	-1.8	-1.8	-1.9		-2.0	-2.3	
Contribution from real exchange rate depreciation	-5.8			-1.4	-1.1	-0.7	-0.8	-0.9	0.2		...	...	
Other identified debt-creating flows	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	8.4			2.3	1.6	1.4	1.5	1.5	1.0		1.0	1.0	
NPV of public sector debt	87.9			87.1	88.9	91.0	93.2	94.9	97.0		106.5	113.9	
o/w foreign-currency denominated	65.6			63.1	62.9	62.9	62.9	62.7	62.5		61.5	52.6	
o/w external	65.6			63.1	62.9	62.9	62.9	62.7	62.5		61.5	52.6	
NPV of contingent liabilities (not included in public sector debt)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need[2]	5.9			7.3	7.9	8.1	8.1	8.1	8.1		8.4	8.7	
NPV of public sector debt-to-revenue ratio (in percent)[3]	520.1			545.9	557.1	570.7	584.1	595.0	608.3		667.5	713.9	
o/w external	387.9			395.5	394.0	394.1	394.3	393.3	392.0		385.4	329.5	
Debt service-to-revenue ratio (in percent)[3,4]	15.4			19.1	20.8	20.8	21.0	20.7	20.7		22.8	25.0	
Primary deficit that stabilizes the debt-to-GDP ratio	0.6			2.3	1.9	1.7	1.7	1.9	1.5		1.7	2.1	
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	2.0	1.1	2.5	2.1	1.7	1.7	1.7	1.7	1.7	1.8	1.7	1.7	1.7
Average nominal interest rate on forex debt (in percent)	1.1	1.8	0.6	1.0	1.3	1.2	1.2	1.1	1.1	1.2	1.1	1.0	1.0
Average real interest rate on domestic currency debt (in percent)	-1.9	-0.9	6.9	-1.2	-0.5	-0.2	0.0	-0.3	-0.3	-0.4	-0.3	-0.2	-0.3
Real exchange rate depreciation (in percent, + indicates depreciatior	-7.4	-0.2	10.6	-1.8	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.1	2.7	6.2	2.0	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Growth of real primary spending (deflated by GDP deflator, in percer	12.5	4.1	16.6	2.2	3.3	2.7	1.7	1.7	1.7	2.2	1.7	1.7	1.7
Grant element of new external borrowing (in percent)	31.6	31.6	0.0	31.6	31.6	31.6	31.6	31.6	31.6	31.6	31.6	31.6	...

Sources: Togolese authorities; and Fund staff estimates and projections.

[1] General government debt.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues excluding grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Togo: External Debt Sustainability Framework, No-Reform Scenario, 2007-2027 [1]
(In percent of GDP, unless otherwise indicated)

	Actual			Projections									
	2006	Historical Average[6]	Standard Deviation[6]	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
External debt (nominal)[1]	80.5			79.3	78.9	78.8	78.7	78.6	78.6		78.6	80.2	
o/w public and publicly guaranteed (PPG)	80.5			79.3	78.9	78.8	78.7	78.6	78.6		78.6	80.2	
Change in external debt	-11.8			-1.3	-0.4	-0.1	-0.1	-0.1	-0.1		0.1	0.2	
Identified net debt-creating flows	-5.7			2.4	1.9	2.4	3.0	3.4	3.4		3.4	3.4	
Non-interest current account deficit	5.2	4.7	2.5	5.7	5.2	5.8	6.4	6.8	6.8		6.8	6.8	6.8
Deficit in balance of goods and services	19.6			21.0	20.8	21.0	21.3	21.3	21.3		21.3	21.3	
Exports	40.5			40.9	40.4	39.9	39.4	39.4	39.4		39.4	39.4	
Imports	60.1			62.0	61.2	60.9	60.7	60.7	60.7		60.7	60.7	
Net current transfers (negative = inflow)	-14.7	-10.2	2.7	-16.0	-15.6	-15.2	-14.8	-14.4	-14.4		-14.4	-14.4	-14.4
Other current account flows (negative = net inflow)	0.3			0.7	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Net FDI (negative = inflow)	-3.4	-2.6	1.0	-2.5	-3.0	-3.0	-3.0	-3.0	-3.0		-3.0	-3.0	-3.0
Endogenous debt dynamics[2]	-7.4			-0.8	-0.3	-0.3	-0.4	-0.4	-0.5		-0.5	-0.4	
Contribution from nominal interest rate	0.9			0.8	1.0	0.9	0.9	0.9	0.8		0.8	0.9	
Contribution from real GDP growth	-1.7			-1.6	-1.3	-1.3	-1.3	-1.3	-1.3		-1.3	-1.3	
Contribution from price and exchange rate changes	-6.6			…	…	…	…	…	…		…	…	
Residual (3-4)[3]	-6.1			-3.6	-2.2	-2.5	-3.1	-3.5	-3.4		-3.3	-3.2	
o/w exceptional financing	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt[4]	65.6			63.1	62.9	62.9	62.9	62.7	62.5		61.5	52.6	
In percent of exports	162.1			154.1	155.4	157.4	159.5	159.1	158.6		155.9	133.3	
NPV of PPG external debt	65.6			63.1	62.9	62.9	62.9	62.7	62.5		61.5	52.6	
In percent of exports	162.1			154.1	155.4	157.4	159.5	159.1	158.6		155.9	133.3	
Debt service-to-exports ratio (in percent)	7.1			7.0	7.5	7.3	7.2	7.3	7.2		7.7	8.3	
PPG debt service-to-exports ratio (in percent)	7.1			7.0	7.5	7.3	7.2	7.3	7.2		7.7	8.3	
Total gross financing need (billions of U.S. dollars)	0.1			0.3	0.1	0.2	0.3	-0.2	-0.2		-0.3	-0.4	
Non-interest current account deficit that stabilizes debt ratio	17.0			6.9	5.5	5.8	6.5	6.9	6.9		6.8	6.6	
Key macroeconomic assumptions													
Real GDP growth (in percent)	2.0	1.1	2.5	2.1	1.7	1.7	1.7	1.7	1.7	1.8	1.7	1.7	1.7
GDP deflator in US dollar terms (change in percent)	7.7	3.8	11.4	5.2	3.8	3.1	3.0	3.2	2.4	3.4	1.8	1.8	1.8
Effective interest rate (percent)[5]	1.0	1.0	0.8	1.0	1.3	1.3	1.2	1.1	1.1	1.2	1.1	1.1	1.1
Growth of exports of G&S (US dollar terms, in percent)	5.1	7.6	14.0	8.7	4.3	3.5	3.4	4.9	4.2	4.8	3.5	3.5	3.5
Growth of imports of G&S (US dollar terms, in percent)	9.9	8.0	11.3	10.8	4.3	4.3	4.3	4.9	4.2	5.5	3.5	3.5	3.5
Grant element of new public sector borrowing (in percent)	...	...	...	35.6	35.6	35.6	35.6	35.6	35.6	35.6	35.6	35.6	35.6
Memorandym item:													
Nominal GDP (billions of US dollars)	2.2			2.4	2.5	2.6	2.8	2.9	3.0		3.6	5.1	

Source: Staff simulations.

[1] Public sector external debt.
[2] Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
[3] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
[4] Assumes that NPV of private sector debt is equivalent to its face value.
[5] Current-year interest payments devided by previous period debt stock.
[6] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 5. Togo: Public Sector Debt Sustainability Framework, Reform Scenario, 2006-27
(Percent of GDP, unless otherwise indicated)

	Actual			Projections									
	2006	Historical Average[5]	Standard Deviation[5]	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt[1]	102.9			93.2	87.0	81.7	53.5	50.7	48.4		40.9	31.3	
of which: foreign-currency denominated	80.5			72.3	68.2	65.0	38.8	37.3	36.0		29.6	19.8	
Change in public sector debt	1.3			-9.7	-6.2	-5.3	-28.2	-2.8	-2.3		-1.2	-0.7	
Identified debt-creating flows	-7.1			-3.7	-3.5	-3.5	-3.6	-1.9	-1.3		-0.7	-0.3	
Primary deficit	1.9	-0.1	2.4	1.6	1.3	1.0	0.7	0.6	0.6	1.0	0.6	0.6	0.6
Revenue and grants	18.3			19.0	20.2	21.3	22.3	22.4	22.4		22.4	22.4	
of which: grants	1.4			2.6	3.2	3.7	4.3	4.2	4.2		4.2	4.2	
Primary (noninterest) expenditure	20.2			20.7	21.5	22.3	23.0	23.0	23.0		23.0	23.0	
Automatic debt dynamics	-8.9			-5.3	-4.8	-4.5	-4.3	-2.5	-1.9		-1.3	-0.9	
Contribution from interest rate/growth differential	-2.3			-3.2	-3.0	-3.1	-2.9	-1.6	-1.5		-1.0	-0.7	
of which: contribution from average real interest rate	-0.3			-0.3	0.2	0.2	0.2	0.5	0.4		0.5	0.4	
of which: contribution from real GDP growth	-2.0			-2.9	-3.2	-3.3	-3.1	-2.1	-2.0		-1.4	-1.1	
Contribution from real exchange rate depreciation	-6.6			-2.1	-1.8	-1.4	-1.4	-0.9	-0.4		...	...	
Other identified debt-creating flows	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	8.4			-6.0	-2.7	-1.8	-24.6	-0.9	-1.1		-0.5	-0.4	
NPV of public sector debt	87.9			66.9	64.8	62.6	35.3	33.4	32.0		29.5	27.0	
of which: foreign-currency denominated	65.6			46.0	46.0	45.8	20.5	19.9	19.5		18.1	15.5	
of which: external	65.6			46.0	46.0	45.8	20.5	19.9	19.5		18.1	15.5	
NPV of contingent liabilities (not included in public sector debt)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need[2]	5.9			7.2	6.7	6.5	6.7	6.7	6.7		6.8	6.9	
NPV of public sector debt-to-revenue ratio (in percent)[3]	520.1			406.7	381.1	355.4	196.0	183.2	175.6		162.1	148.3	
of which: external	387.9			279.7	270.5	260.4	114.0	109.6	107.2		99.7	85.3	
Debt service-to-revenue ratio (in percent) [3,4]	15.4			18.3	13.2	10.4	9.6	10.4	10.2		10.9	11.6	
Primary deficit that stabilizes the debt-to-GDP ratio	0.6			11.3	7.5	6.3	28.9	3.4	2.9		1.8	1.3	
Key macroeconomic and fiscal assumptions													
Real GDP growth (percent)	2.0	1.1	2.5	2.9	3.5	4.0	4.0	4.0	4.0	3.7	3.5	3.5	3.6
Average nominal interest rate on forex debt (percent)	1.1	1.8	0.6	1.0	0.5	0.5	0.4	0.7	0.6	0.6	0.7	1.3	0.9
Average real interest rate on domestic currency debt (percent)	-1.9	-0.9	6.9	-1.2	2.7	3.0	3.4	3.9	4.3	2.7	4.7	3.0	4.1
Real exchange rate depreciation (percent, "+" indicates depreciation)	-8.3	-1.2	10.5	-2.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, percent)	2.1	2.7	6.2	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Growth of real primary spending (deflated by GDP deflator, percent)	12.5	4.1	16.6	5.5	7.6	7.9	7.3	4.0	4.0	6.0	3.5	3.5	3.6
Grant element of new external borrowing (percent)	31.6	31.6	0.0	31.6	31.6	31.6	31.6	31.6	31.6	31.6	31.6	31.6	...

Sources: Togolese authorities; and Fund staff estimates and projections.
[1] General government debt.
[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
[3] Revenues excluding grants.
[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.
[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 6. Togo: Sensitivity Analysis for Key Indicators of Public Debt (Reform Scenario), 2007-27

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Reform Scenario	67	65	63	35	33	32	30	27
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	...	...	...	35	34	34	38	53
A2. Primary balance reverts to 2007 level from 2011	...	...	...	35	34	34	38	51
A3. Permanently lower GDP growth[1]	...	...	...	35	34	33	33	42
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviations in 2011-12	...	...	...	35	36	37	36	37
B2. Primary balance at historical average minus one standard deviations in 2011-12	...	...	...	35	35	35	34	33
B3. Combination of B1-B2 using one-half standard deviation shocks	...	...	...	35	35	36	35	35
B4. One-time 30 percent nominal depreciation in 2011	...	...	...	35	42	38	34	33
B5. 10 percent of GDP increase in other debt-creating flows in 2011	...	...	...	35	43	42	42	46
NPV of Debt-to-Revenue Ratio[2]								
Reform Scenario	351	321	294	158	149	143	132	120
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	...	...	...	196	189	188	208	291
A2. Primary balance reverts to 2007 level from 2011	...	...	...	196	189	187	207	282
A3. Permanently lower GDP growth[1]	...	...	...	196	185	179	179	229
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviations in 2011-12	...	...	...	196	197	206	199	203
B2. Primary balance at historical average minus one standard deviations in 2011-12	...	...	...	196	192	194	185	182
B3. Combination of B1-B2 using one-half standard deviation shocks	...	...	...	196	194	200	192	192
B4. One-time 30 percent nominal depreciation in 2011	...	...	...	196	230	209	186	184
B5. 10 percent of GDP increase in other debt-creating flows in 2011	...	...	...	196	238	233	232	251
Debt Service-to-Revenue Ratio[2]								
Reform Scenario	16	11	9	8	8	8	9	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	...	...	...	10	10	10	13	20
A2. Primary balance reverts to 2007 level from 2011	...	...	...	10	10	10	12	19
A3. Permanently lower GDP growth[1]	...	...	...	10	10	10	11	16
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviations in 2011-12	...	...	...	10	10	10	13	15
B2. Primary balance at historical average minus one standard deviations in 2011-12	...	...	...	10	10	10	12	14
B3. Combination of B1-B2 using one-half standard deviation shocks	...	...	...	10	10	10	13	15
B4. One-time 30 percent nominal depreciation in 2011	...	...	...	10	13	12	13	14
B5. 10 percent of GDP increase in other debt-creating flows in 2011	...	...	...	10	10	11	15	19

Sources: Togolese authorities; and Fund staff estimates and projections.

[1] Assumes that real GDP growth is is lowered by one percentage point over the length of the projection period.

[2] Revenues are defined excluding grants.

Table 7. Togo: External Debt Sustainability Framework, Reform Scenario, 2006-27
(Percent of GDP, unless otherwise indicated)

	Actual			Projections									
	2006	Historical Average[5]	Standard Deviation[5]	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
External public debt (nominal)	80.5			72.3	68.2	65.0	38.8	37.3	36.0		29.6	19.8	
of which: public and publicly guaranteed (PPG)	80.5			72.3	68.2	65.0	38.8	37.3	36.0		29.6	19.8	
Change in external debt	-4.3			-8.2	-4.1	-3.2	-26.2	-1.4	-1.4		-1.1	-0.8	
Identified net debt-creating flows	-5.0			1.8	0.7	-0.3	-1.2	-0.5	-0.8		-1.0	-0.9	
Non-interest current account deficit	5.2	4.7	2.5	5.6	6.0	5.9	5.5	5.2	4.9		4.3	4.1	4.3
Deficit in balance of goods and services	19.6			20.9	21.5	21.5	21.3	20.6	19.9		18.6	18.6	
Exports	40.5			40.6	41.7	42.8	43.9	45.0	46.1		48.3	48.3	
Imports	60.1			61.5	63.2	64.3	65.2	65.6	66.0		66.9	66.9	
Net current transfers (negative = inflow)	-14.7	-10.3	2.8	-15.9	-16.1	-16.2	-16.4	-15.9	-15.5		-14.7	-14.8	-14.7
Other current account flows (negative = net inflow)	0.3			0.7	0.6	0.6	0.6	0.5	0.5		0.4	0.2	
Net FDI (negative = inflow)	-3.4	-2.6	1.0	-2.5	-3.3	-4.0	-4.5	-4.5	-4.5		-4.5	-4.5	-4.5
Endogenous debt dynamics[1]	-6.8			-1.4	-2.0	-2.2	-2.2	-1.2	-1.2		-0.8	-0.4	
Contribution from nominal interest rate	0.9			0.8	0.3	0.3	0.3	0.3	0.2		0.2	0.3	
Contribution from real GDP growth	-1.5			-2.2	-2.4	-2.5	-2.4	-1.4	-1.4		-1.0	-0.7	
Contribution from price and exchange rate changes	-6.1			…	…	…	…	…	…		…	…	
Residual [2]	0.7			-10.0	-4.8	-2.9	-25.0	-0.9	-0.6		-0.1	0.1	
of which: exceptional financing	0.0			-6.3	0.0	0.0	-18.8	0.0	0.0		0.0	0.0	
NPV of external debt [3]	65.6			46.0	46.0	45.8	20.5	19.9	19.5		18.1	15.5	
(percent of exports)	162.1			113.3	110.2	107.1	46.7	44.3	42.3		37.6	32.1	
NPV of PPG external debt	65.6			46.0	46.0	45.8	20.5	19.9	19.5		18.1	15.5	
(percent of exports)	162.1			113.3	110.2	107.1	46.7	44.3	42.3		37.6	32.1	
Debt service-to-exports ratio (percent)	7.1			7.0	3.2	2.2	2.0	2.4	2.3		2.6	3.2	
PPG debt service-to-exports ratio (percent)	7.1			7.0	3.2	2.2	2.0	2.4	2.3		2.6	3.2	
Total gross financing need (billions of U.S. dollars)	0.1			0.3	0.1	0.2	0.2	-0.4	-0.4		-0.5	-0.7	
Non-interest current account deficit that stabilizes debt ratio	9.5			13.9	10.1	9.2	31.7	6.6	6.3		5.4	4.9	
Key macroeconomic assumptions													
Real GDP growth (percent)	2.0	0.8	2.5	2.9	3.5	4.0	4.0	4.0	4.0	3.7	3.5	3.5	3.6
GDP deflator in US dollar terms (annual change, percent)	7.7	4.6	11.8	5.2	4.0	3.3	3.2	3.4	2.6	3.6	2.0	2.0	2.0
Effective interest rate (percent) [4]	1.1	1.2	0.4	1.0	0.5	0.5	0.4	0.7	0.6	0.6	0.7	1.3	0.9
Growth of exports of G&S (U.S. dollar terms, percent)	5.1	9.4	13.5	8.7	10.5	10.3	10.0	10.2	9.3	9.8	5.6	5.6	6.0
Growth of imports of G&S (U.S. dollar terms, percent)	9.9	8.4	11.9	10.8	10.6	9.4	8.8	8.1	7.5	9.2	5.6	5.6	5.8
Grant element of new public sector borrowing (percent)	...	...	...	34.1	34.1	34.1	34.1	34.1	34.1	34.1	34.1	34.1	34.1
Memorandum item:													
Nominal GDP (billions of US dollars)	2.2			2.4	2.6	2.8	3.0	3.2	3.4		4.5	7.8	

Source: Staff simulations.

[1] Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
[2] Includes exceptional financing (i.e., changes in arrears and debt relief),changes in gross foreign assets, valuation adjustments, and (for projections only) the contribution from price and exchange rate changes.
[3] Assumes that NPV of private sector debt is equivalent to its face value.
[4] Current-year interest payments divided by previous period debt stock.
[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 8. Togo: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt (Reform Scenario), 2007-27
(Percent)

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of debt-to-GDP ratio								
Reform scenario	46	46	46	21	20	20	18	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 [1]	...	...	...	21	22	23	29	39
A2. New public sector loans on less favorable terms in 2008-27 [2]	...	...	...	23	22	22	23	23
B. Bound Tests								
B1. Real GDP growth reduced by 1 percentage point relative to baseline from 2011	...	...	...	21	20	20	20	21
B2. Export value growth at historical average minus one standard deviation in 2011-12 [3]	...	...	...	21	21	20	19	16
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2011-12	...	...	...	21	23	25	24	20
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2011-12 [4]	...	...	...	21	29	35	33	22
B5. Combination of B1-B4 using one-half size shocks	...	...	...	22	26	31	29	24
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 [5]	...	...	...	21	28	26	23	22
NPV of debt-to-exports ratio								
Reform scenario	113	110	107	47	44	42	38	32
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 [1]	...	...	...	47	48	49	61	81
A2. New public sector loans on less favorable terms in 2008-27 [2]	...	...	...	52	50	48	47	48
B. Bound Tests								
B1. Real GDP growth reduced by 1 percentage point relative to baseline from 2011	...	...	...	47	45	43	41	43
B2. Export value growth at historical average minus one standard deviation in 2011-12 [3]	...	...	...	49	54	61	54	46
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2011-12	...	...	...	49	46	44	39	33
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2011-12 [4]	...	...	...	49	63	77	68	46
B5. Combination of B1-B4 using one-half size shocks	...	...	...	50	60	70	63	52
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 [5]	...	...	...	49	75	72	63	44
Debt service-to-exports ratio								
Reform scenario	7	3	2	2	2	2	3	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 [1]	...	...	...	2	3	3	3	6
A2. New public sector loans on less favorable terms in 2008-27 [2]	...	...	...	3	3	3	3	4
B. Bound Tests								
B1. Real GDP growth reduced by 1 percentage point relative to baseline from 2011	...	...	...	2	2	2	3	4
B2. Export value growth at historical average minus one standard deviation in 2011-12 [3]	...	...	...	2	3	4	4	4
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2011-12	...	...	...	2	3	3	3	3
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2011-12 [4]	...	...	...	2	3	3	3	4
B5. Combination of B1-B4 using one-half size shocks	...	...	...	2	3	4	4	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 [5]	...	...	...	3	3	4	3	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) [6]	32	32	32	32	32	32	32	32

Source: Staff projections and simulations.

[1] Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flo

[2] Assumes that the interest rate on new borrowing is 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the

[3] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level a (implicitly assuming an offsetting adjustment in import levels).

[4] Includes official and private transfers and FDI.

[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

[6] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.